Dropping a Quantum Temporary

Supplementary material

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The proofs the paper defers and the catalogue it summarises. This document is not self-contained and is not meant to be read alone.

A Deferred proofs

The proofs in this appendix are the ones the body states in one sentence each; nothing in them is used before it is stated. The conventions are the body's. A map K from the visible register to the joint register has *Kraus operators* $K_i = (\langle i| \otimes I)K$ and is *drop-safe* when $K = g \otimes W$, equivalently when the K_i are pairwise proportional; the zero map is drop-safe. A record is *reachable* for a command when the selected denotation along it is nonzero. The three programs of the body are recalled first, since the proofs are checked on them.

Example A.1 (the running examples).

(a) *superpose, then drop*	(b) *superpose, then read*	(c) *measure, then read*
`let t = new |0>`	`let t = new |0>`	`let t = new |0>`
`  t := H t;`	`  t := H t;`	`  t := tilt t;`
		`  let p = lift(meas t in S)`
	`  v := v + t;`	`  v := v + t;`
`  drop t; v`	`  drop t; v`	`  drop t; v`

A.1 The lifted calculus in full, and its metatheory

The body displays the grammar and three reductions of the command calculus. Here are the rest. A command is typed in a *parameter count* n, the number of parameters bound so far, and a *flag* $\ell \in \{\mathsf{live}, \mathsf{dropped}\}$ for the temporary; the parameter bound by the n-th lift is written p_n, and a boolean expression e is built from parameters, negation and conjunction. The rules are

$$\frac{}{n; \ell \vdash \mathtt{halt}}\ \textsc{Halt} \qquad \frac{U^\top U = I \quad n; \mathsf{live} \vdash c}{n; \mathsf{live} \vdash \mathtt{gate}\, U\, ; c}\ \textsc{Gate} \qquad \frac{V^\top V = I \quad n; \ell \vdash c}{n; \ell \vdash \mathtt{vgate}\, V\, ; c}\ \textsc{VGate}$$

$$\frac{n; \mathsf{dropped} \vdash c}{n; \mathsf{live} \vdash \mathtt{drop}\, t\, ; c}\ \textsc{Drop} \qquad \frac{S \subseteq [d] \quad n+1; \mathsf{live} \vdash c}{n; \mathsf{live} \vdash \mathtt{let}\ p_n = \mathtt{lift}(\mathtt{meas}\, t \in S)\ \mathtt{in}\ c}\ \textsc{Lift}$$

$$\frac{\mathrm{params}(e) \subseteq \{p_0, \ldots, p_{n-1}\} \quad n; \ell \vdash c_1 \quad n; \ell \vdash c_2}{n; \ell \vdash \mathtt{if}\ e\ \mathtt{then}\ c_1\ \mathtt{else}\ c_2}\ \textsc{If}$$

A *command* is typed when $0; \mathsf{live} \vdash c$. Only Gate, Lift and Drop name the temporary, and all three need it live; so in a typed command nothing names the temporary after its drop. The typed program of the body's affine calculus is the lift-free command $\mathtt{gate}\, U_1; \ldots; \mathtt{gate}\, U_k; \mathtt{drop}\, t; \mathtt{halt}$ that its elaboration lists.

The machine. Fix a measurement record o, an infinite sequence over $\{\mathsf{in}, \mathsf{out}\}$; the machine reads it from the left. A configuration $\langle c; \psi; \eta; \sigma\rangle$ carries the command, the joint state $\psi \in \mathbb{R}^d \otimes \mathbb{R}^m$, a store η from parameters to booleans, and the unread suffix σ of o. With $P_{\mathsf{in}} = P_S$, $P_{\mathsf{out}} = I - P_S$, $\beta(\mathsf{in}) = 1$ and $\beta(\mathsf{out}) = 0$, the reductions are

$$\begin{aligned}
\langle \texttt{gate}\ U\,; c; \psi; \eta; \sigma\rangle &\longrightarrow \langle c; U\psi; \eta; \sigma\rangle,\\
\langle \texttt{vgate}\ V\,; c; \psi; \eta; \sigma\rangle &\longrightarrow \langle c; (I \otimes V)\psi; \eta; \sigma\rangle,\\
\langle \texttt{drop}\ t; c; \psi; \eta; \sigma\rangle &\longrightarrow \langle c; \psi; \eta; \sigma\rangle,\\
\langle \texttt{let}\ \ p_n = \texttt{lift}(\texttt{meas}\ t \in S)\ \texttt{in}\ c; \psi; \eta; b \cdot \sigma\rangle &\longrightarrow \langle c; (P_b \otimes I)\psi; \eta[p_n \mapsto \beta(b)]; \sigma\rangle,\\
\langle \texttt{if}\ e\ \texttt{then}\ c_1\ \texttt{else}\ c_2; \psi; \eta; \sigma\rangle &\longrightarrow \langle c_1; \psi; \eta; \sigma\rangle \text{ if } \eta \vDash e,\ \text{else } \langle c_2; \psi; \eta; \sigma\rangle,
\end{aligned}$$

and a configuration whose command is `halt` is terminal. A configuration is *well typed* when $n; \ell \vdash c$ and η binds exactly $p_0, \dots, p_{n-1}$. The initial configuration at visible input v is $\langle c; e_{i_0} \otimes v; \emptyset; o\rangle$.

The selected denotation. By recursion on c, with a store η and a record σ:

$$\begin{aligned}
E_{\sigma,\eta}(\texttt{halt}) &= I, \qquad E_{\sigma,\eta}(\texttt{gate}\ U\,; c) = E_{\sigma,\eta}(c)\,U,\\
E_{\sigma,\eta}(\texttt{vgate}\ V\,; c) &= E_{\sigma,\eta}(c)\,(I \otimes V), \qquad E_{\sigma,\eta}(\texttt{drop}\ t\,; c) = E_{\sigma,\eta}(c),\\
E_{b\cdot\sigma,\eta}(\texttt{let}\ \ p_n &= \texttt{lift}(\texttt{meas}\ t \in S)\ \texttt{in}\ c)\\
&= E_{\sigma,\eta[p_n \mapsto \beta(b)]}(c)\,(P_b \otimes I),\\
E_{\sigma,\eta}(\texttt{if}\ e\ \texttt{then}\ c_1\ \texttt{else}\ c_2) &= E_{\sigma,\eta}(c_1) \text{ if } \eta \vDash e,\ \text{else } E_{\sigma,\eta}(c_2).
\end{aligned}$$

The selected denotation of a typed command along o is $E_o(c)(v) = E_{o,\emptyset}(c)(e_{i_0} \otimes v)$. The drop of c is *operationally safe* under o when the map from visible inputs to terminal joint states factors as $g \otimes W v$.

Theorem A.2 (metatheory of the lifted calculus). *Fix a measurement record o.*

(1) A well-typed configuration is terminal or steps; the step preserves typing, the parameter store binds exactly the parameters in scope, and no step names a dropped temporary.

(2) Every run halts, and its terminal joint state is $E_o(c)$ applied to the visible input, where $E_o(c)$ is the selected denotation of c along o.

Hence the drop of c is operationally safe under o if and only if $E_o(c)$ is drop-safe.

Proof. Two parts, each a standard pair.

(1) Progress and preservation. Progress. Let $\langle c; \psi; \eta; \sigma\rangle$ be well typed with $c \neq \texttt{halt}$.

- a gate, a visible gate or a drop steps unconditionally;
- a lift steps because σ is a suffix of the infinite record o and so has a head;
- a conditional steps because $\eta \vDash e$ is defined: If confines the parameters of e to $p_0, \dots, p_{n-1}$, which η binds.

Preservation. The gate, visible-gate, drop and conditional steps keep n and η and continue with a premise of the rule that typed c, at the same n; the lift step continues with a command typed at $n + 1$ and extends η by exactly p_n. So the store binds exactly the parameters in scope after every step. After `drop` t the continuation is typed with the flag dropped, and no rule with that flag in its conclusion names the temporary; so no later step names it.

(2) Termination and adequacy. Termination. Every step replaces the command by a proper subcommand, and commands are finite trees.

Adequacy. By induction on the length of the run: if $\langle c;\psi;\eta;\sigma\rangle$ runs to $\langle \mathtt{halt};\psi';\eta';\sigma'\rangle$ then $\psi' = E_{\sigma,\eta}(c)\,\psi$. For a terminal configuration $E_{\sigma,\eta}(\mathtt{halt}) = I$. Otherwise the first step is one of the five reductions, each matching one clause of the recursion:

- a gate contributes U, and a visible gate $I \otimes V$;
- a drop contributes nothing;
- a lift on outcome b contributes the projection $P_b \otimes I$, together with the extended store;
- a conditional selects the branch that η selects.

The induction hypothesis applied to the continuation finishes. Starting from $\langle c; e_{i_0} \otimes v; \emptyset; o\rangle$, the terminal state is $E_{o,\emptyset}(c)(e_{i_0} \otimes v) = E_o(c)(v)$.

The last sentence. It is a definition read twice: operational drop safety asks whether $v \mapsto E_o(c)(v)$ factors, which is drop safety of $E_o(c)$. □

The proportionality lemma. The body's channel theorem and its contextual-equivalence theorem both reduce to one linear-algebra fact.

Lemma A.3 (proportionality). *Let $K_0, \dots, K_{d-1}$ and M be linear maps on $\mathbb{R}^m$, and suppose $\sum_i (K_i x)(K_i x)^\top = (Mx)(Mx)^\top$ for every x. Then $K_i = c_i M$ for scalars c_i.*

Proof. Write k_{ib} and m_b for the b-th rows. The (b,b) entry of the hypothesis reads $\sum_i \langle k_{ib}, x\rangle^2 = \langle m_b, x\rangle^2$ for all x; taking $x \perp m_b$ forces every $\langle k_{ib}, x\rangle = 0$, so $k_{ib} \in (m_b^\perp)^\perp = \operatorname{span}(m_b)$: $k_{ib} = c_{ib} m_b$, with $\sum_i c_{ib}^2 = 1$ when $m_b \neq 0$. For two nonzero rows $m_b, m_{b'}$ the (b,b') entry gives $\sum_i c_{ib} c_{ib'} \langle m_b, x\rangle\langle m_{b'}, x\rangle = \langle m_b, x\rangle\langle m_{b'}, x\rangle$; the product of two nonzero linear forms is not the zero polynomial, so $\sum_i c_{ib} c_{ib'} = 1$, and Cauchy–Schwarz between the unit vectors $(c_{ib})_i$ and $(c_{ib'})_i$ holds with equality only when they coincide. So $c_{ib} = c_i$ does not depend on b over the nonzero rows of M, and a zero row of M is a zero row of every K_i; hence $K_i = c_i M$. □

Contextual indistinguishability. The body's theorem that the criterion is contextual indistinguishability.

Theorem A.4 (the criterion is contextual indistinguishability). *Let F be any class of visible gates containing the reflections. A typed program is weakenable if and only if F does not distinguish it from some separable program.*

Proof. ($\Rightarrow$) A weakenable program has the denotation of a separable one, so no continuation distinguishes them. ($\Leftarrow$) Let $\operatorname{sep}(R, W)$ be undistinguished, with $g = Re_{i_0}$ a unit vector, so its Kraus operators are $g_i W$ with $\sum_i g_i^2 = 1$. For a unit $u \neq e_0$, the reflection exchanging e_0 and u is in F and satisfies $C^\top e_0 = u$, so agreement of the two distributions at $b = 0$ says $\sum_i \langle u, K_i x\rangle^2 = \langle u, Wx\rangle^2$ for all such u and all x, and both sides are continuous in u, so it holds at e_0 as well. These are the quadratic forms of the symmetric matrices $\sum_i (K_i x)(K_i x)^\top$ and $(Wx)(Wx)^\top$, which therefore coincide, and lemma A.3 gives $K_i = c_i W$. So the program is drop-safe, hence weakenable by the body's criterion theorem. □

The summed channel. The body's remark that an analysis reading the channel a program implements cannot decide the drop rests on one pair of commands.

Proposition A.5 (the summed channel does not decide). *There are two well-typed commands on a temporary qubit and a visible qubit, each with two reachable records, that have the same summed channel $\Phi_0 + \Phi_1$ and the same weight on every branch at every input, and opposite verdicts on every branch.*

Proof. Both apply H to the temporary, reaching $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes v$. The first then applies CZ, lifts on $\{0\}$, and drops: CZ gives $\frac{1}{\sqrt{2}}(|0\rangle \otimes v + |1\rangle \otimes Zv)$, so the branch selecting 0 has denotation $\frac{1}{\sqrt{2}}|0\rangle \otimes I$ and the other $\frac{1}{\sqrt{2}}|1\rangle \otimes Z$, one Kraus operator each: both drops are safe, with branch maps $X \mapsto X/2$ and $X \mapsto ZXZ/2$. The second lifts on $\{0\}$, applies $t := t \oplus v$ (the visible bit controlling), and drops: the branch selecting b keeps $\frac{1}{\sqrt{2}}|b\rangle \otimes v$, and the controlled write sends it to $\frac{1}{\sqrt{2}} \sum_c v_c \, |b \oplus c\rangle \otimes |c\rangle$, whose Kraus operators are $\frac{1}{\sqrt{2}} P_0$ and $\frac{1}{\sqrt{2}} P_1$ in some order, not proportional: both drops are unsafe, with branch map $X \mapsto \mathrm{diag}(X)/2$ twice. Both pairs sum to the dephasing channel $X \mapsto \mathrm{diag}(X)$, and every branch weight is $\mathrm{tr}(X)/2$. □

A product at every input. The body's remark that "a product at every input" admits programs drop safety excludes is the following dichotomy.

Proposition A.6 (a product at every input). *Let $K : \mathbb{R}^m \to \mathbb{R}^d \otimes \mathbb{R}^m$ be linear, and suppose Kv is a product vector, or zero, for every v. Then K has one of exactly two shapes:*

the temporary is fixed $K = g \otimes W$ for a vector $g \in \mathbb{R}^d$ and a matrix W on $\mathbb{R}^m$ — the visible register varies, the temporary does not;

the visible register is fixed $Kv = Gv \otimes w$ for a fixed nonzero $w \in \mathbb{R}^m$ and a linear $G : \mathbb{R}^m \to \mathbb{R}^d$ — the temporary varies, the visible register does not.

The first shape is drop safety. A map of the second is drop-safe exactly when G has rank at most one, so such an isometry with $m \geq 2$ never is.

Proof. For each v with $Kv \neq 0$ write $Kv = g_v \otimes w_v$ with g_v and w_v nonzero. Either all the g_v are parallel or two of them are not, and the two cases are settled separately.

All the temporary factors on one line. If every g_v lies on the line of one vector g, put $W = (\langle g| \otimes I)\, K / \|g\|^2$. For $Kv = g_v \otimes w_v = \lambda_v \, g \otimes w_v$ this gives $Wv = \lambda_v w_v$ and $Kv = g \otimes Wv$; for $Kv = 0$ both sides vanish. So $K = g \otimes W$.

Otherwise the visible factors are. Choose v_1, v_2 with g_{v_1}, g_{v_2} linearly independent. By linearity

$$K(v_1 + v_2) \;=\; g_{v_1} \otimes w_{v_1} + g_{v_2} \otimes w_{v_2},$$

which as a $d \times m$ matrix is $g_{v_1} w_{v_1}^\top + g_{v_2} w_{v_2}^\top$, of rank two unless w_{v_1} and w_{v_2} are parallel. A product vector, or zero, has rank at most one, so they are parallel to one w. Any v_3 with $Kv_3 \neq 0$ has g_{v_3} independent of g_{v_1} or of g_{v_2}, since otherwise g_{v_1} and g_{v_2} would be parallel, and the same argument gives w_{v_3} parallel to w. So every w_v lies on the line of w, and $G = (I \otimes \langle w|)\, K / \|w\|^2$ is linear with $Kv = Gv \otimes w$ for every v, as before.

The last sentence. $K_i = (\langle i| \otimes I)K$ sends v to $(Gv)_i \, w$, so $K_i = w \, e_i^\top G$, and these are pairwise proportional exactly when the rows of G are, that is, when G has rank at most one. If K is an isometry then $\|Gv\| \, \|w\| = \|v\|$ for every v, so G is injective and has rank m. □

Without typing. The body's remark that the norm clause is the type system's whole contribution to the criterion.

Proposition A.7 (without typing). *For a program of the grammar whose gates are arbitrary linear maps, weakenability is equivalent to drop safety together with $\|Kv\| = \|v\|$ for every v. The norm clause cannot be dropped: at $d = m = 1$ the one-gate program that doubles the visible register is drop-safe and not weakenable.*

Proof. The criterion theorem's proof uses the isometry hypothesis at one step, where it makes W orthogonal, and the norm clause supplies exactly that step. It cannot be dropped, because a witness is a typed program and so denotes an isometry: at $d = m = 1$ every map has a single Kraus operator and is therefore drop-safe, while $v \mapsto 2v$ is not an isometry and no typed program has its denotation. □

What the Gram matrix is. Written over $\mathbb{C}$; over $\mathbb{R}$ read $\dagger$ as $\top$ and $\bar{U}$ as U, and drop the second half of (1).

Proposition A.8 (what the Gram matrix is). *Let K be the denotation of a typed program, with Kraus operators K_i and Gram matrix Γ.*

(1) The drop is safe if and only if $K_i^\dagger K_j = c_{ij} I$ for all i, j, and then $c = \Gamma/m$; over $\mathbb{C}$, if and only if the temporary's final state is the same at every unit input.

(2) $\operatorname{rank}\Gamma$ is the least temporary dimension of a typed program whose visible channel $\rho \mapsto \sum_i K_i \rho K_i^\dagger$ is that of K.

(3) Γ is unchanged by a visible gate after the scope, and a gate U on the temporary carries it to $\bar{U}\Gamma U^\top$.

Proof. All three parts are statements about one family: the Kraus operators $K_1, \dots, K_d$ of K, and the matrix Γ they determine.

(1) Safety is the scalar condition. Two directions, then the reading over $\mathbb{C}$.

Safe $\Rightarrow$ scalar. If $K = g \otimes W$ with $\|g\| = 1$ then W is an isometry of the visible register, hence unitary, and $K_i^\dagger K_j = \bar{g}_i g_j I$.

Scalar $\Rightarrow$ safe. Let $K_i^\dagger K_j = c_{ij} I$. Taking traces, $m\, c_{ij} = \Gamma_{ij}$, so c is Hermitian and positive semidefinite; write $c = VDV^\dagger$ with V unitary and D diagonal, and put $U = V^\top$. Rotating the temporary by U diagonalises the family:

$$K'_k = \sum_i U_{ki} K_i, \qquad K_k'^\dagger K'_l = \sum_{ij} \bar{U}_{ki} U_{lj} c_{ij} I = (\bar{U} c U^\top)_{kl} I = (V^\dagger c V)_{kl} I = D_{kl} I.$$

At most one diagonal entry is nonzero: if λ_k and λ_l were both nonzero then $K'_k/\sqrt{\lambda_k}$ and $K'_l/\sqrt{\lambda_l}$ would be isometries of $\mathbb{C}^m$, hence unitaries, with product $K_k'^\dagger K'_l = 0$, which is impossible. So $K'_k = 0$ for all k but one, k_0, and

$$K = (U^\dagger \otimes I) K' = (U^\dagger e_{k_0}) \otimes K'_{k_0}$$

is drop-safe.

The state reading, and why it is complex. The temporary's final state at a unit input v has entries

$$\rho_t(v)_{ij} = \langle K_j v, K_i v \rangle = v^\dagger K_j^\dagger K_i v.$$

Over $\mathbb{C}$ a matrix whose quadratic form $v \mapsto v^\dagger A v$ is constant on unit vectors is a scalar, so ρ_t is input-independent exactly when every $K_j^\dagger K_i$ is a scalar, which is the first half. Over $\mathbb{R}$ this fails. With J the rotation of $\mathbb{R}^2$ by a right angle, take

$$K \;=\; \tfrac{1}{\sqrt{2}}\big(|0\rangle \otimes I + |1\rangle \otimes J\big).$$

Since $v^\top J v = 0$, it leaves the temporary in $\frac{1}{2}I$ at every real unit input, while I and J are not proportional; at the complex input $\frac{1}{\sqrt{2}}(1, i)$ the temporary is pure.

(2) The rank is the least temporary dimension. Let X be the $d \times m^2$ matrix whose rows are the K_i written out. Then $X X^\dagger = \Gamma^\top$, so

$$\operatorname{rank}\Gamma \;=\; \operatorname{rank} X \;=\; \dim \operatorname{span}\{K_i\},$$

which is the rank of the channel's Choi matrix and, as the two bounds below show, the least number of Kraus operators in any representation of it—its *Choi rank* [11, p. 78].

- *No smaller dimension works.* A typed program with a temporary of dimension d' gives its channel d' Kraus operators, so $d' \geq \operatorname{rank}\Gamma$.
- *That dimension is achieved.* Let $r = \operatorname{rank}\Gamma$ and let $K'_1, \ldots, K'_r$ be Kraus operators of the channel, which exist because a channel always has a Kraus family of exactly that many [4, 11]. The channel is trace preserving because K is an isometry, so

  $$V v \;=\; \sum_k |k\rangle \otimes K'_k v$$

  is an isometry $\mathbb{C}^m \to \mathbb{C}^r \otimes \mathbb{C}^m$. Its columns extend to a unitary U on $\mathbb{C}^r \otimes \mathbb{C}^m$ with $U(e_0 \otimes v) = V v$, and the program that allocates at e_0, applies U and drops has denotation V, temporary dimension r, and the channel of K.

(3) How the Gram matrix transports. A visible gate C after the scope replaces K_i by CK_i, and $\operatorname{tr}(K_i^\dagger C^\dagger C K_j) = \Gamma_{ij}$. A gate U on the temporary replaces K_i by $\sum_j U_{ij} K_j$, and the Gram matrix of that family is $\bar{U}\Gamma U^\top$, as computed in (1). □

The Gram matrix of a family of vectors is the standard invariant of the family in quantum information: it decides linear independence, hence unambiguous discrimination [3], and it is the data of the family up to a unitary. Part (1) is the Knill–Laflamme condition [7] for the code that is the whole visible register, with the K_i as the errors and the temporary as the environment; its rank-one coefficient matrix says the correction is trivial, and the information-disturbance tradeoff of Kretschmann, Schlingemann and Werner [8] is its quantitative form, a channel being close to correctable exactly when its complementary channel is close to constant. The second half of (1) is noninterference in the sense of Goguen and Meseguer [5], with the visible input as the high input and the temporary as the low observer; Ying, Feng and Yu define a quantitative version for quantum automata by the distance between the observer's states along purged and unpurged runs [12], and the impurity ι of section A.5 is that degree for the drop.

A.2 Clifford+T splits, in every dimension

The relativised criterion. The body states the criterion over a gate ring and defers its proof here.

Theorem A.9 (the relativised criterion). *Let $\mathcal{A}$ be a gate ring completable in the temporary's dimension and U a program over $\mathcal{A}$. Then U is weakenable by gates over $\mathcal{A}$ if and only if U is drop-safe and $\mathcal{A}$ splits* $\operatorname{proj}(U)$.

Proof. First the object both halves are about. If U is drop-safe with unit final state g, then $U_{(i,b),(0,c)} = g_i W_{bc}$, and row 0 of the unitary W is a unit vector, so $\mathrm{proj}(U) = gg^\dagger$.

A witness makes the projector split. A witness $\mathrm{sep}(R, W)$ over $\mathcal{A}$ denotes $g \otimes W$ with $g = Re_0$, a column of a unitary over $\mathcal{A}$. So U is drop-safe and $\mathrm{proj}(U) = gg^\dagger$ splits, with $z = g$.

A splitting vector builds a witness. Let $gg^\dagger = zz^\dagger$ with z over $\mathcal{A}$. Then z is a unit vector on the line of g, say $z = \lambda g$ with $|\lambda| = 1$, and the witness is assembled in two pieces.

- *The temporary factor.* By completability z is a column of a unitary over $\mathcal{A}$; composing with a permutation matrix, which is over $\mathcal{A}$, gives a unitary R over $\mathcal{A}$ with $Re_0 = z$. At $d = 2$ take $R = \left(\begin{smallmatrix} z_0 & -\bar{z}_1 \\ z_1 & \bar{z}_0 \end{smallmatrix}\right)$, over $\mathcal{A}$ because $\mathcal{A}$ is closed under conjugation: every gate ring is completable at $d = 2$.
- *The visible factor.* Put

$$W'_{bc} \;=\; \sum_i \bar{z}_i U_{(i,b),(0,c)} \;=\; \langle z, g\rangle W_{bc} \;=\; \bar{\lambda} W_{bc},$$

  which is $\bar{\lambda}W$ and so unitary, and which lies over $\mathcal{A}$ because it is a polynomial in entries of U and of $\bar{z}$.

Then $\mathrm{sep}(R, W')$ is typed, over $\mathcal{A}$, and denotes $z \otimes \bar{\lambda}W = g \otimes W$, the denotation of U. □

Write $\mathcal{R} = \mathbb{Z}[1/\sqrt{2}, i] = \mathbb{Z}[\omega][1/\sqrt{2}]$ with $\omega = e^{i\pi/4}$, and $\mathcal{R}_0 = \mathbb{Z}[1/\sqrt{2}]$ for its real elements. Three facts about these rings are used. First, $\mathcal{R}$ is a principal ideal domain: $\mathbb{Z}[\omega]$ is the ring of integers of $\mathbb{Q}(\zeta_8)$, whose class number is one (a classical fact; the body gives the reference), and a localisation of a principal ideal domain is one. Second, the units of $\mathcal{R}_0$ are $\pm\sqrt{2}^{\,a}(1+\sqrt{2})^b$ with $a, b \in \mathbb{Z}$: the units of $\mathbb{Z}[\sqrt{2}]$ are $\pm(1+\sqrt{2})^b$, and inverting the prime $\sqrt{2}$ adds the one generator. Third, three unit norms: $|\sqrt{2}|^2 = 2$, $|1+\sqrt{2}|^2 = (1+\sqrt{2})^2$, and $|1+\omega|^2 = 2+\omega+\bar{\omega} = 2+\sqrt{2} = \sqrt{2}(1+\sqrt{2})$, where $1+\omega$ is a unit of $\mathcal{R}$ because $(2+\sqrt{2})(2-\sqrt{2}) = 2$ and $2 = (\sqrt{2})^2$ is a unit. Write σ for the automorphism of $\mathbb{Q}(\omega)$ with $\sigma(\sqrt{2}) = -\sqrt{2}$; the Galois group is abelian, so σ commutes with complex conjugation and $\sigma(|x|^2) = |\sigma(x)|^2 \geq 0$.

Theorem A.10 (the Clifford+T coefficient ring splits). *Let $\mathcal{R} = \mathbb{Z}[1/\sqrt{2}, i]$ and $d \geq 1$.*

(1) Every rank-one projection on $\mathbb{C}^d$ with entries in $\mathcal{R}$ is $zz^\dagger$ for a vector z over $\mathcal{R}$.

(2) Consequently, at a temporary of any dimension and any visible register, a program over $\mathcal{R}$ is weakenable by gates over $\mathcal{R}$ if and only if it is drop-safe.

Proof. Part (2) is part (1) fed to the relativised criterion, so the work is in (1).

(1) Every rank-one projection over $\mathcal{R}$ splits. Let $P = gg^\dagger$ with $\|g\| = 1$ have entries in $\mathcal{R}$. The splitting vector is built in four steps.

(1) *A column, made primitive.* Some column $r = Pe_j = \bar{g}_j\, g$ is nonzero; it lies in $\mathcal{R}^d$, and $P = rr^\dagger/\|r\|^2$. Divide r by a greatest common divisor of its entries, which exists because $\mathcal{R}$ is a principal ideal domain: $r = \delta r'$ with r' primitive, and still $P = r'r'^\dagger/N$ for

$$N \;=\; \|r'\|^2 \;=\; \textstyle\sum_i |r'_i|^2 \;\in\; \mathcal{R}_0.$$

(2) *The norm is a unit.* $NP_{ik} = r'_i \bar{r}'_k$, so N divides every product $r'_i \bar{r}'_k$; these products generate the ideal $(r'_0, \ldots, r'_{d-1})(\bar{r}'_0, \ldots, \bar{r}'_{d-1}) = (1)$. So N is a unit of $\mathcal{R}$ and, being real, of $\mathcal{R}_0$: $N = \sqrt{2}^{\,a}(1+\sqrt{2})^b$, positive.

(3) *The exponents have equal parity.* The conjugate $\sigma(N) = \sum_i |\sigma(r_i')|^2$ is positive too, and $\sigma(N) = (-\sqrt{2})^a(1-\sqrt{2})^b$ has the sign of $(-1)^{a+b}$; so $a \equiv b \pmod 2$.

(4) *The unit whose norm is N.* Put $c = a \bmod 2$ and

$$\varepsilon \;=\; \sqrt{2}^{\,(a-c)/2}\,(1+\sqrt{2})^{(b-c)/2}\,(1+\omega)^c,$$

a unit of $\mathcal{R}$ with $|\varepsilon|^2 = \sqrt{2}^{\,a-c}(1+\sqrt{2})^{b-c}\big(\sqrt{2}(1+\sqrt{2})\big)^c = N$.

Then $z = r'/\varepsilon$ has entries in $\mathcal{R}$, $\|z\|^2 = N/|\varepsilon|^2 = 1$, and $zz^\dagger = r'r'^\dagger/N = P$. At $d = 1$ the four steps degenerate rather than fail: the only rank-one projection on $\mathbb{C}^1$ is $P = (1)$, so $r' = 1$, $N = 1$, $a = b = c = 0$ and $\varepsilon = 1$, giving $z = 1$.

(2) So the criterion is unchanged over $\mathcal{R}$. A drop-safe program over $\mathcal{R}$ has $\mathrm{proj}(U) = gg^\dagger$ over $\mathcal{R}$, which splits by (1), and $\mathcal{R}$ is completable in every dimension by the column lemma of Giles and Selinger cited in the body; the body's relativised criterion then gives a witness over $\mathcal{R}$. The converse is the criterion's other direction. □

Proposition A.11 (the restricted rings). *Over $\mathbb{Z}[1/2, i]$ and over $\mathbb{Z}[i/\sqrt{2}]$, every rank-one projection with entries in the ring, in every dimension, is $zz^\dagger$ for a z over the ring. Over $\mathbb{Z}[1/2]$ the projection onto $|+\rangle$ is not, and over $\mathbb{Z}[1/\sqrt{2}]$ the projection onto $\cos\frac{\pi}{8}|0\rangle + \sin\frac{\pi}{8}|1\rangle$ is not.*

Proof. The proof of (1) used only three things about $\mathcal{R}$: that it is a principal ideal domain closed under conjugation, that its real elements form $\mathcal{R}_0$, and that every totally positive unit of $\mathcal{R}_0$ is $|\varepsilon|^2$ for a unit ε of $\mathcal{R}$. So the two positive cases are a matter of checking those three.

The two rings that split.

- $\mathbb{Z}[1/2, i] = \mathbb{Z}[i][1/2]$. $\mathbb{Z}[i]$ has class number one, the real elements form $\mathbb{Z}[1/2]$, whose positive units are the 2^a, and $2^a = |(1+i)^a|^2$ with $1+i$ a unit since $2 = -i(1+i)^2$.
- $\mathbb{Z}[i/\sqrt{2}] = \mathbb{Z}[\sqrt{-2}][1/2]$. $\mathbb{Z}[\sqrt{-2}]$ has class number one, the real elements again form $\mathbb{Z}[1/2]$, and $2^a = |\sqrt{-2}^{\,a}|^2$ with $\sqrt{-2}$ a unit since $2 = -(\sqrt{-2})^2$.

The three class numbers are one because the Minkowski bound of each of $\mathbb{Q}(i)$ and $\mathbb{Q}(\sqrt{-2})$ is below 2, and that of $\mathbb{Q}(\zeta_8)$ is below 3 while its only prime of norm 2 is $(1-\omega)$. So the argument of (1) goes through unchanged.

The two that do not. Over a real ring $\mathcal{A}$ a splitting vector z of a real projection $P = gg^\top$ is λg with $\lambda \in \mathcal{A}$ and $|\lambda| = 1$, so $z = \pm g$. That is what fails:

- over $\mathbb{Z}[1/2]$, the projection $P = \frac{1}{2}\left(\begin{smallmatrix}1&1\\1&1\end{smallmatrix}\right)$ has entries in the ring and $\pm(1,1)/\sqrt{2}$ does not;
- over $\mathbb{Z}[1/\sqrt{2}]$, the projection $\frac{1}{2}(I + (X+Z)/\sqrt{2})$ has entries in the ring, and no unit vector on its ray does, by the computation below.

□

The magic state, and the program that realises the separation. Take g, the H-type magic state $\cos\frac{\pi}{8}|0\rangle + \sin\frac{\pi}{8}|1\rangle$, whose ray is $[(1+\sqrt{2},\ 1)]$ and whose projector $\frac{1}{2}(I + (X+Z)/\sqrt{2})$ has entries in $\mathbb{Z}[1/\sqrt{2}]$. A unit vector on that ray is $\lambda(1+\sqrt{2}, 1)$ with $\lambda^2 = (2-\sqrt{2})/4$, whose norm down to $\mathbb{Q}$ is $1/8$, not a rational square, so no such λ lies in $\mathbb{Q}(\sqrt{2})$, let alone in $\mathbb{Z}[1/\sqrt{2}]$, and the projector does not split. Over $\mathcal{R}$ it does, with $\omega = e^{i\pi/4}$: $z = \left(\frac{1+\omega}{2}, \frac{i(1-\omega)}{2}\right)$ has $zz^\dagger = \frac{1}{2}\left(I + \frac{X+Z}{\sqrt{2}}\right)$. One program realises the separation: the single gate $U = rot \otimes rot$, where rot is the rotation of a qubit

by $\pi/8$. Its entries are $\cos^2\frac{\pi}{8}=\frac{2+\sqrt{2}}{4}$, $\cos\frac{\pi}{8}\sin\frac{\pi}{8}=\frac{\sqrt{2}}{4}$ and $\sin^2\frac{\pi}{8}=\frac{2-\sqrt{2}}{4}$, all in $\mathbb{Z}[1/\sqrt{2}]$, though rot alone is over neither ring, and $U(|0\rangle\otimes v)=g\otimes rot\,v$, so the drop is safe with the magic state as its final state. Over $\mathcal{R}$ this program is weakenable, contracting against $\bar{z}$ for the visible factor $e^{-i\pi/8}rot$; over $\mathbb{Z}[1/\sqrt{2}]$ it is not.

A.3 A command whose verdict depends on the outcome

Planned The proofs for the body's boundary section, and then the smallest temporary that still obstructs, the fragments that escape, a family forcing a verdict to read many record entries, and the closed form for consecutive lifts. Being written; nothing here is claimed.

A.4 Several temporaries

Planned The proofs for the body's section on several temporaries, and then what a sound procedure summary may forget, and how many test states certify a borrowed temporary. Being written; nothing here is claimed.

A.5 The price of the verdict

Throughout, a *circuit program* is a program of the calculus whose temporary and visible registers are qubits, $d=2^k$ and $m=2^n$, the temporary allocated at $|0^k\rangle$, whose gates act on at most three qubits of the joint register and are given by their matrices, each entry an algebraic number presented exactly by its minimal polynomial and an isolating rectangle in binary; its size counts the gates, the qubits and that presentation, and scalars are complex. K is the denotation, an isometry $\mathbb{C}^m\to\mathbb{C}^d\otimes\mathbb{C}^m$ with Kraus operators $K_i=(\langle i|\otimes I)K$. For a unit vector ψ of the visible register tensored with a reference register R of any finite dimension, $\rho_t(\psi)=\mathrm{Tr}_{vR}\big[(K\otimes I_R)\,\psi\psi^\dagger\,(K\otimes I_R)^\dagger\big]$ is the state of the temporary after the run; its *impurity* $1-\mathrm{tr}\,\rho_t(\psi)^2$ vanishes exactly when it is pure, $\iota(K)=\sup_\psi\big(1-\mathrm{tr}\,\rho_t(\psi)^2\big)$ is its worst case, and $\bar{\iota}(K)$ is its value at the maximally entangled input $\psi=m^{-1/2}\sum_c|c\rangle\otimes|c\rangle$, whose visible marginal is the maximally mixed state. The reference is essential: the swap $Kv=v\otimes e_0$ of the body's productwise proposition leaves a pure temporary at every pure input, yet at the maximally entangled input its temporary is maximally mixed and $\iota(K)=\bar{\iota}(K)=1-1/m$. Since $\rho_t(\psi)=\mathrm{Tr}_v[K\sigma K^\dagger]$ depends on ψ only through its visible marginal σ, a reference of dimension m suffices, the supremum is attained, and $\bar{\iota}(K)\le\iota(K)$. Promise problems are meant throughout: membership is in the promise version of the class, and hardness is under polynomial-time many-one reductions of promise problems.

The temporary's state at the maximally entangled input. It is $\Gamma^\top/m$; hence $\bar{\iota}(K)=1-\mathrm{tr}(\Gamma^2)/m^2$, and K is drop-safe if and only if $\bar{\iota}(K)=0$, if and only if $\iota(K)=0$. Both propositions below rest on this.

The joint state at the maximally entangled input is $(K\otimes I_R)\psi$, and tracing out R leaves $KK^\dagger/m$ on temporary and visible; tracing out the visible register then gives $\mathrm{Tr}_v[KK^\dagger]/m$. Its (i,j) entry is $\sum_{a,c}K_{(i,a),c}\,\overline{K_{(j,a),c}}=\sum_{a,c}(K_i)_{ac}\overline{(K_j)_{ac}}=\mathrm{tr}(K_j^\dagger K_i)=\Gamma_{ji}$, so the state is $\Gamma^\top/m$; its trace is $\sum_i\mathrm{tr}(K_i^\dagger K_i)=\mathrm{tr}(K^\dagger K)=m$ divided by m, as it must be, and its purity is $\mathrm{tr}(\Gamma^2)/m^2$. For the equivalences: a state of trace one is pure exactly when it is a rank-one projector, so $\bar{\iota}(K)=0$ says Γ has rank one, which is form (4) of the body's definition of drop safety. If $K=g\otimes W$ with unit g then for every input, with or without a reference, the joint state is $g\otimes(W\otimes I_R)\psi$ and $\rho_t(\psi)=gg^\dagger$, so $\iota(K)=0$; and $\iota(K)=0$ implies $\bar{\iota}(K)=0$, which closes the cycle. (Since Γ is positive semidefinite

with trace m, rank one is also the identity $\Gamma^2 = m\Gamma$; the checker takes the rank, and the suite checks the two against each other.)

Theorem A.12 (the classical price). *Deciding whether a circuit program over the gate set* $\{X, \text{CNOT}, \text{Toffoli}\}$ *is drop-safe is coNP-complete.*

Proof. Every gate in $\{X, \text{CNOT}, \text{Toffoli}\}$ permutes the joint computational basis, so a program over them does too. Write its denotation as

$$Ke_x \;=\; |a(x)\rangle \otimes |b(x)\rangle,$$

where x ranges over visible basis inputs, $a(x)$ is the basis state the temporary is left in and $b(x)$ the one the visible register is left in. By the body's definition the drop is safe exactly when the temporary factor is one vector for every x, that is, exactly when a *is constant.* Both halves below are about that one function.

Membership. A pair $x \neq y$ with $a(x) \neq a(y)$ is a certificate of unsafety, and it is polynomially checkable. These programs contain no measurement, so a program is one list of gates applied in order, with no branching and nothing to select. A joint basis state is a bit string — the visible positions holding x, the allocated ones their initial values — and each of the three gates reads at most three positions of that string and rewrites one: X flips its own bit, CNOT flips its target when the control bit is 1, and Toffoli flips its target when both control bits are 1. Each gate is therefore a constant-time update of the string, so for a program of s gates $a(x)$ — the temporary's positions once all s have been applied — costs $O(s)$. Safety is the absence of such a pair, so the problem is in coNP.

Hardness. We reduce from unsatisfiability. The drop sees constancy of a, and what must be decided is unsatisfiability, so the reduction takes three steps: evaluate B reversibly, read a off the resulting program, and convert constancy into unsatisfiability with one evaluation. Figure A.1 draws the reduction on a two-gate B.

Let B be a Boolean circuit over $\{\wedge, \neg\}$ with $n \geq 1$ inputs and s gates; a CNF formula converts to such a circuit in polynomial size. Taking $n \geq 1$ is free: a formula with no variables is decided by one evaluation, and the visible register below is the one carrying B's inputs, so it must be nonempty for the fixed unsafe program of step 3 to exist.

(1) *Evaluate B reversibly*, by Bennett's compute–copy–uncompute [1]. Allocate s scratch qubits and one target, all at $|0\rangle$, and in topological order replace each gate of B by a reversible one writing into a *fresh* scratch qubit:

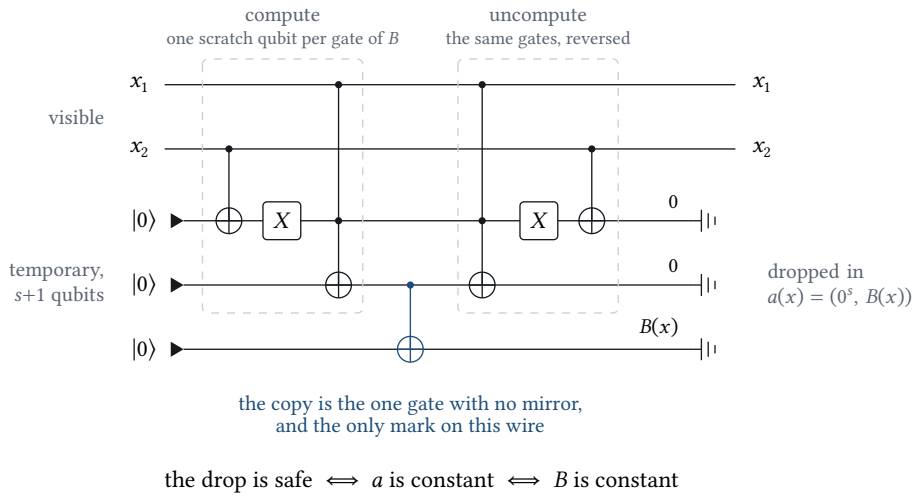
- $\wedge$ with inputs on wires u, w becomes a Toffoli from u, w into the fresh scratch qubit;
- $\neg$ with input on wire u becomes a CNOT from u into the fresh scratch qubit, followed by X on it.

Then a CNOT copies the circuit's output wire into the target, and then the whole gate sequence is applied again in reverse order. Every gate used is an involution, so the reverse pass returns each scratch qubit to $|0\rangle$ on every input x; only the target keeps a value. The net map is

$$|x\rangle \otimes |0^s, 0\rangle \;\longmapsto\; |x\rangle \otimes |0^s, B(x)\rangle.$$

(2) *Read the verdict off it.* Take the temporary to be the register of $s+1$ qubits (the scratch and the target), allocated at $|0^{s+1}\rangle$, and the visible register to be x. This is a circuit program of size $O(n+s)$, and by step 1 its denotation is

$$Ke_x \;=\; |0^s, B(x)\rangle \otimes e_x, \qquad \text{so} \qquad a(x) = (0^s,\ B(x)).$$



Drawn for $B = x_1 \wedge \neg x_2$, so $s = 2$. Constancy is not yet unsatisfiability, so evaluate $B(0^n)$ first: emit this program when $B(0^n) = 0$, a fixed unsafe one when $B(0^n) = 1$. Either way the output is drop-safe exactly when B is unsatisfiable.

Fig. A.1. Look at the mirror. Every gate drawn permutes the computational basis, so the temporary is left in one basis state $a(x)$ for each input, and every gate of the compute block has a partner in the uncompute block, so the scratch qubits reach the drop at $|0\rangle$ on every input. The copy into the target is the one gate without a partner, and the only mark on the target's wire; that is why the temporary reaches the drop holding $B(x)$ and nothing else, and why the drop is safe exactly when B is constant.

So a is constant exactly when B is, and the drop is safe if and only if B is constant.

(3) *Convert constancy into unsatisfiability.* A constant B may be constantly 1, so constancy is not yet unsatisfiability. One evaluation closes the gap. Compute $B(0^n)$ in polynomial time and output:

- if $B(0^n) = 1$: a fixed unsafe program—a CNOT from the first visible qubit into a temporary qubit. Here B is satisfiable, and the output is unsafe.
- if $B(0^n) = 0$: the program of step 2. It is safe exactly when B is constant, and $B(0^n) = 0$ makes that constant 0; so it is safe exactly when $B \equiv 0$, that is, when B is unsatisfiable.

In both cases the program produced is drop-safe if and only if B is unsatisfiable, and the reduction runs in polynomial time. So the problem is coNP-hard, and with the first half, coNP-complete. □

With the $s+1$ temporaries kept as separate names in the calculus of the body's section on several temporaries, the same thing holds per name: each scratch qubit is separately constant and the target is separately constant exactly when B is constant. The predicate is not the one of Liu, Zhou and Meng [9], who ask whether some unitary can restore the ancilla while preserving the data and prove that coNP-complete over {Toffoli, X} by a reduction from satisfiability through Jordan's strong equivalence of reversible circuits [6]. The two predicates separate on the program above: it stores $B(x)$ in the target, so it is uncomputable in their sense for every B, by running backwards, and drop-safe in ours only for constant B.

The exact verdict, with no promise on the size of the leak, inherits theorem A.12 over any gate set containing X, CNOT and Toffoli, and lies in PSPACE for gates over a fixed number field, since each entry of K is a sum of exponentially many products of gate entries, computable in polynomial

space, and form (3) of the body's definition enumerates the minors; Tanaka shows the exact form of non-identity check to be NQP-complete [10], and where the exact drop verdict sits between coNP and PSPACE is open.

Proposition A.13 (the price at the maximally mixed input). *Fix a polynomial $p : \mathbb{N} \to \mathbb{N}$ with $p(n) \geq 1$ for every n, let $\varepsilon = 1/p(\mathit{size})$, and let the circuit program's gates come from a fixed finite set. Deciding, for such a program promised either drop-safe or $\bar{\iota}(K) \geq \varepsilon$, which holds is in BQP.*

Proof. The algorithm estimates one number, $\operatorname{tr} \rho^2$ for $\rho = \Gamma^\top/m$, and reads the promise off it.

The algorithm. A uniformly random basis input e_x leaves the temporary in $\operatorname{Tr}_v[Ke_xe_x^\dagger K^\dagger]$, whose average over x is

$$\operatorname{Tr}_v[KK^\dagger]/m \;=\; \Gamma^\top/m \;=\; \rho,$$

the temporary's state at the maximally entangled input; two runs on independently drawn inputs leave the two temporaries in $\rho \otimes \rho$. The swap test on two registers in state $\rho \otimes \sigma$ accepts with probability $\frac{1}{2}(1 + \operatorname{tr}(\rho\sigma))$ [2], here

$$\tfrac{1}{2}(1 + \operatorname{tr} \rho^2) \;=\; 1 - \tfrac{1}{2}\bar{\iota}(K).$$

Repeat $O(1/\varepsilon^2)$ times and accept when the empirical acceptance rate exceeds $1-\varepsilon/4$: by Hoeffding's inequality the two cases, rate 1 and rate at most $1 - \varepsilon/2$, are told apart with constant error. Every step runs the program—whose gates the machine applies directly, which is why the fixed finite gate set is hypothesised—or a swap test on polynomially many qubits. For gates presented as arbitrary algebraic matrices the bound needs a compilation step with its own error analysis, which we do not carry out.

The reflection named in the body. For $m \geq 2$, $U = I - 2|w\rangle\langle w|$ has nonzero $+1$ and -1 eigenspaces, and the two readings of the leak come apart:

- an input superposing a $+1$ with a -1 eigenvector makes $c = \langle\psi|(U \otimes I)|\psi\rangle$ vanish, the temporary maximally mixed and $\iota(K_U) = \frac{1}{2}$;
- at the maximally mixed input $c = \operatorname{tr}(U)/m = 1 - 2/m$, so $\bar{\iota}(K_U) = \frac{1}{2}(1 - (1 - 2/m)^2) = 2/m - 2/m^2$.

□

Planned The gate sets on which the verdict is cheap, and what a gate one step outside such a set costs. Being written; the statement is not in the body either.

B The catalogue

B.1 Weakenable without being separable: three witnesses

The body's *weakening witness* asks that *some* typed program of the shape $\operatorname{sep}(R, W)$ have the same denotation, not that the program be of that shape. The difference is easy to state and easy to lose, so here are three catalogue programs beside the witness the body's criterion produces for each. The middle column is the denotation the two share.

the program	**its denotation**	**a witness** $\mathrm{sep}(R, W)$
`silq-two-gate`: `t := H t; t := H t; drop t`	$\lvert 0\rangle \otimes v$	$R = I, W = I$ — allocate and drop, touching nothing
`dj-constant`: the body's Deutsch–Jozsa text, with its joint U_f	$\lvert -\rangle \otimes HD_fHv$	$R = HX$, carrying $\lvert 0\rangle$ to $\lvert -\rangle$; $W = HD_fH$, which is $\pm I$
`grover-oracle`: compute f into the ancilla, kick the phase back, uncompute	$\lvert 0\rangle \otimes D_fv$	$R = I; W = D_f = \mathrm{diag}((-1)^{f(x)})$, a legal gate because it is real and orthogonal

None of the three programs *is* $\mathrm{sep}(R, W)$: each has a gate that touches both registers, and the first has two gates on the temporary where the shape allows one. All three are weakenable, and by theorem A.4 no continuation distinguishes any of them from the witness beside it. Two catalogue programs, `in-shape` and `do-nothing`, are literally of the shape; the other ten safe ones are in the position of these three.

Each witness is computed the way the criterion's proof computes it: read g and W off the denotation, then prepare g from e_{i_0} by the reflection through $(e_{i_0} - g)^{\perp}$, which for $g = \lvert -\rangle$ and $e_{i_0} = \lvert 0\rangle$ is HX up to sign.

The seventeen catalogue programs of the body, with the verdict of the criterion and of the two modelled producer rules. **R1** accepts when every gate is a permutation; **R2** adds that no gate writes visible data into the temporary. Computed by `dqt catalogue` and pinned by three tests: one asserting R1 unsound, one asserting R2 sound and strictly incomplete, and one asserting the totals printed here. The names are the checker's own and only `example-B` is one of the body's running programs: it is program (b). Program (a) is `park` and program (c) is `tilt-read`; `example-A` and `example-C` are two further programs, not those.

program	R1	R2	criterion
`example-A`	accept	accept	safe
`example-B`	refuse	refuse	unsafe
`example-C`	refuse	refuse	safe
`park`	refuse	refuse	safe
`silq-two-gate`	refuse	refuse	safe
`tilt-read`	refuse	refuse	unsafe
`in-shape`	refuse	refuse	safe
`dj-constant`	refuse	refuse	safe
`dj-balanced`	refuse	refuse	safe
`bv-0`	refuse	refuse	safe
`bv-1`	refuse	refuse	safe
`grover-oracle`	accept	refuse	safe
`kickback`	refuse	refuse	unsafe
`copy-copy`	refuse	refuse	safe
`cz-copy`	refuse	refuse	unsafe
`compute-alone`	**accept**	refuse	**unsafe**
`do-nothing`	accept	accept	safe
accepts	4	**2**	**12**

Of the twelve safe programs, the ray each leaves the temporary on, and the verdict of the restore-to-allocation rule (`dqt pool`):

program	left on	restore rule
`example-A`, `silq-two-gate`, `grover-oracle`, `do-nothing`	$[1, 0]$	accepts
`example-C`, `park`, `in-shape`, `copy-copy`	$[1, 1]$	**refuses**
`dj-constant`, `dj-balanced`, `bv-0`, `bv-1`	$[1, -1]$	**refuses**

The runtime-check table emitted by `dqt residual` for the witness command of the body's boundary section: on the branch selecting $\{0, 1\}$, REFUSE; on the branch selecting $\{2\}$, ADMIT. On the same command with the read deleted, ADMIT on both.

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